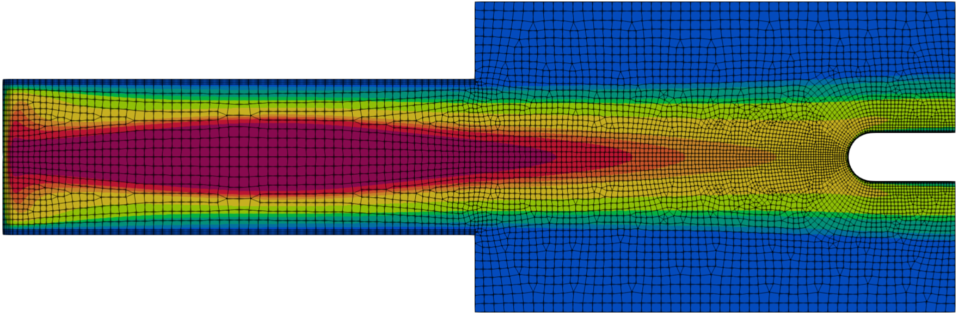
A Monolithic High-Order Multi-Domain Solver for Inductively Coupled Plasma

Nicolas Corthouts



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Introduction

Au fait, qu'est-ce que j'ai fait pendant 4 ans?

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J'ai développé un **solver numérique** pour les **plasmas** à **induction**.

Etats de la matière: eau à pression atmosphérique

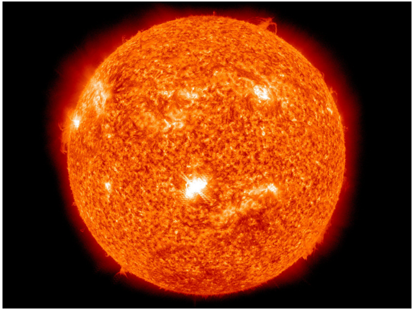
Solide

Liquide

Gaz

Que se passe-t-il si on chauffe encore plus?

Plasma: le quatrième état de la matière



Les plasmas composent 90% de l'univers visible.

Fusion nucléaire, médecine, métallurgie, lasers, création de microprocesseurs, ...

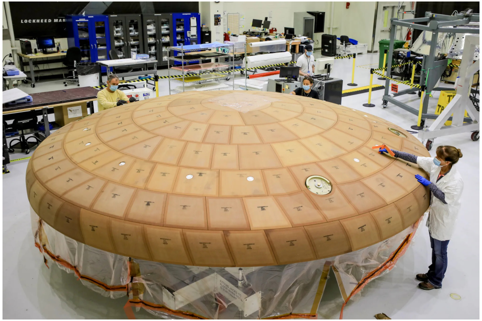
Plasma en réentrée atmosphérique


[1]

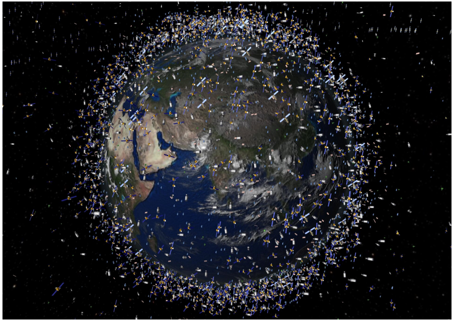
Vitesse de réentrée $\simeq 10\,\mathrm{km\,s^{-1}}$: choc transformant l'air en plasma.

[1] Coutesy of NASA.

Sytème de protection thermique et destruction des déchets spatiaux



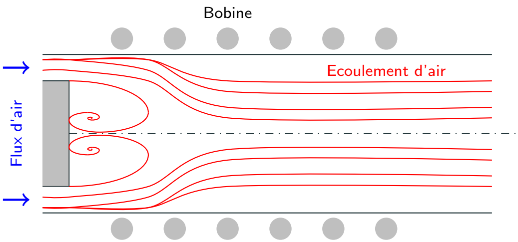
TPS



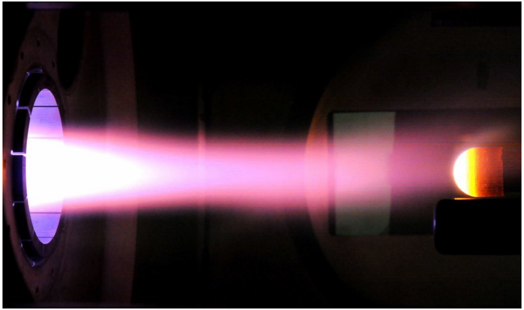
Déchets spatiaux

Expériences reproduisant les plasmas de réentrée atmosphérique.

Plasma à induction



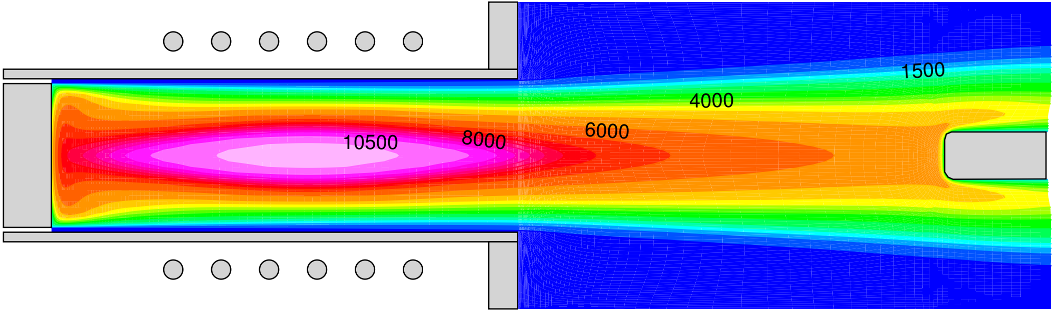
Torche



Chambre

Exemples: Plasmatron (VKI), Plasmatron X (Illinois), IPG (Russie), ...

Plasma à induction: solveurs numériques



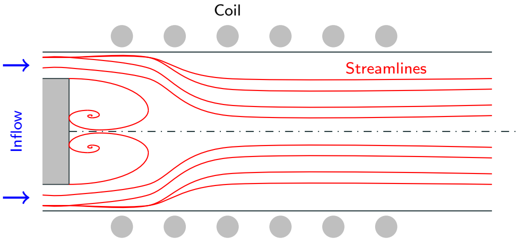
[2]

On propose un nouveau type de solver numérique pour les plasmas à induction.

[2] Courtesy of Thierry Magin.

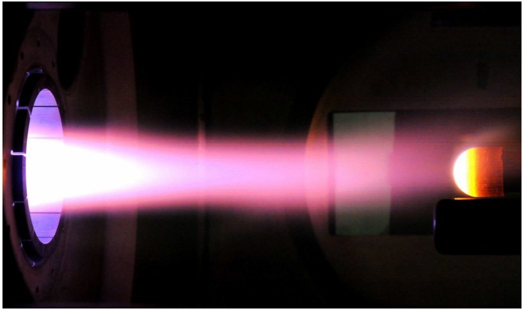
Thesis motivation

ICP facility: an overview



Torch

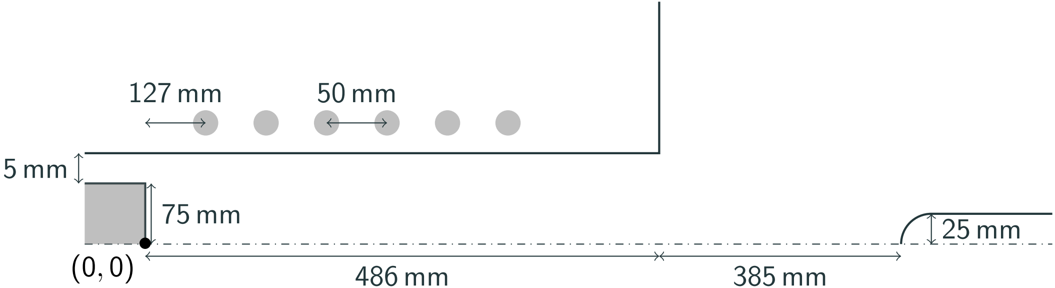
- $T \simeq 10\,000\,\mathrm{K}$, $p \in [1000\,\mathrm{Pa}; 100\,000\,\mathrm{Pa}]$.
- $Re \simeq 100$, $Ma \simeq 0.01$.
- Air, argon, carbon dioxide.



Chamber

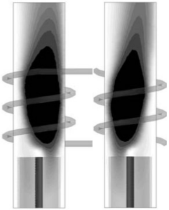
- At LTE & non equilibrium.
- Local heat transfer principle.

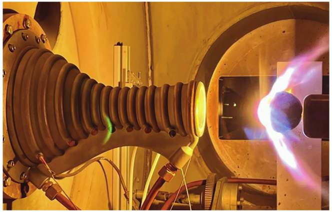
ICP : geometry

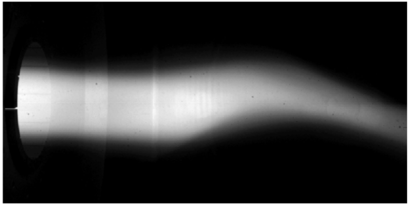


Large temperature gradients!

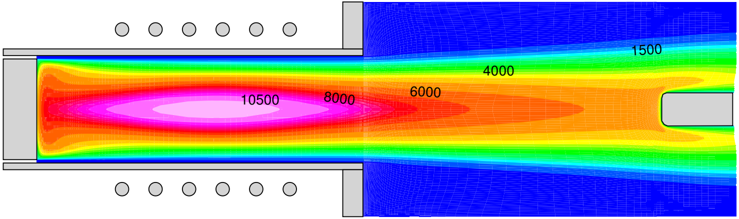
ICP simulations : complex physics at play







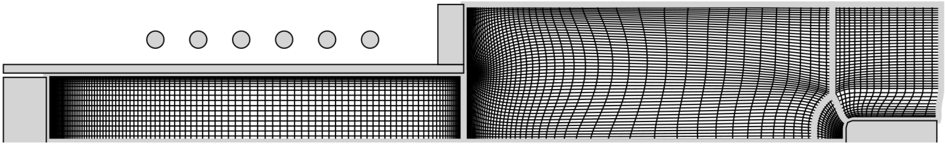
ICP : existing solvers



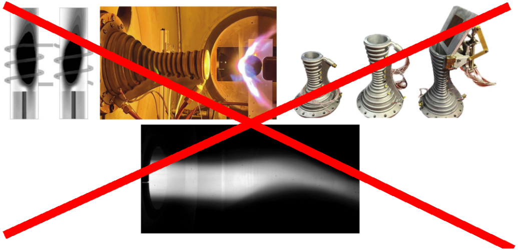
- Axisymmetric.
- Subsonic.
- Steady-state.
- Local thermodynamic equilibrium.
- No-radiation.

- Relatively slow convergence.
- Requires a constant attention during convergence processes.

Finite volume solver limitations.



- Constant solution on each element: refined mesh large gradient region.
- Dissipative, "smoothing" the physics.
- Structured mesh.



Thesis objective

Developing a hybridized discontinuous Galerkin solver of inductively coupled plasmas.

- Solution is not constant anymore on every element.
- Much greater mesh flexibility and (hopefully) less mesh elements.
- We hope to make a more robust solver.
- More precise (less dissipative) solver than finite volumes.

Research questions

Q1 In addition of being precise, can a high-order solver be robust for inductively coupled plasma applications?

Q2 Is the developed solver user-friendly?

Q3 Can the new solver be easily adapted to the new experiments performed in ICP facilities?

Model for inductively coupled plasma

A plasma is a quasi-neutral gas composed of charged and neutral particles exhibiting a collective behaviour (through collisions = particles interaction).

Short ranged

Direct contact (local & binary)

Long ranged

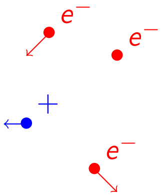
Electric force (distance and collective)

For scales > Debye length (1 µm in our case), $q \simeq 0$ due to electric force.

Cold plasmas and thermal equilibrium

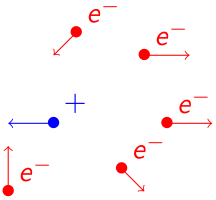
"Cold plasmas" store energy in the free electrons. It is then transmitted during collisions to the ions and neutrals.

Innefficient collisions $(T_e \gg T_h)$



Thermal non-equilibrium

Efficient collisions $(T_e \simeq T_h)$



Thermal equilibrium

Chemical equilibrium state

Collisions may lead to chemical reactions.

$$N + O \leftrightharpoons NO^{+} + e^{-}$$

$\tau_{chem} \gg \tau_{hydro}$	$\tau_{chem} \simeq \tau_{hydro}$	$\tau_{chem} \ll \tau_{hydro}$
Chemical equilibrium	Chemical non-equilibrium	Frozen flow

Thermal + Chemical equilibrium = Local Thermodynamic Equilibrium (LTE)

General equation for unmagnetized, optically thick plasma at LTE

$$
\begin{cases}
\partial_t \rho + \nabla \cdot (\rho \mathbf{v}) & = 0 \\
\partial_t (\rho \mathbf{v}) + \nabla \cdot (\rho \mathbf{v}\mathbf{v} + p\mathbb{I}) - \nabla \cdot \boldsymbol{\tau} & = \mathbf{j} \times \mathbf{B} \\
\partial_t \left(\frac{1}{2} \rho ||\mathbf{v}||^2 + \rho e \right) + \nabla \cdot (\rho H \mathbf{v}) + \nabla \cdot (\mathbf{q} - \boldsymbol{\tau} \cdot \mathbf{v}) & = \mathbf{j} \cdot \mathbf{E}
\end{cases}
$$

$$
\boldsymbol{\tau} = \eta \left[\nabla \mathbf{v} + (\nabla \mathbf{v})^T - \frac{2}{3} (\nabla \cdot \mathbf{v}) \mathbb{I} \right]
$$

$$
\mathbf{q} = -\lambda \nabla T
$$

$$
\mathbf{j} = \sigma \mathbf{E} - \sum_{i,j \in S} n_i q_i D_{ij} \left[\frac{\nabla p_j}{n k_B T_h} - \frac{y_j p}{n k_B T_h} \nabla \ln(p) + \left(k_T^h + k_T^e \right) \nabla \ln(T) \right]
$$

Mutation++

Multicomponent Thermodynamic And Transport properties for IONized gases in C++

Navier-Stokes and Maxwell equations

$$\begin{cases} \partial_t \rho + \nabla \cdot (\rho \mathbf{v}) & = 0 \\ \partial_t (\rho \mathbf{v}) + \nabla \cdot (\rho \mathbf{v}\mathbf{v} + p\mathbb{I}) - \nabla \cdot \boldsymbol{\tau} & = \mathbf{j} \times \mathbf{B} \\ \partial_t \left(\frac{1}{2} \rho ||\mathbf{v}||^2 + \rho e \right) + \nabla \cdot (\rho H \mathbf{v}) + \nabla \cdot (\mathbf{q} - \boldsymbol{\tau} \cdot \mathbf{v}) & = \mathbf{j} \cdot \mathbf{E} \end{cases}$$

$$\nabla \cdot \mathbf{E} = 0 \qquad \partial_t \mathbf{B} + \nabla \times \mathbf{E} = \mathbf{0}$$

$$\nabla \cdot \mathbf{B} = 0 \qquad -\epsilon_0 \partial_t \mathbf{E} + \frac{1}{\mu_0} \nabla \times \mathbf{B} = \mathbf{j}$$

Hypothesis specific to ICP

- Quasi-steady & averaged over an oscillation period.
  $\partial_t f \int_0^{1/f} dt \simeq 0$
- Axisymmetric
  $\partial_\theta \simeq 0$
- No displacement current.
  $\varepsilon_0 \partial_t \mathbf{E} \simeq 0$
- Ambipolar electrostatic field ($j_z = j_r = 0$).
  $\mathbf{j} = \sigma_e \mathbf{E}_I$
- Split of the induced electric field in a coil and plasma part
  $E_I = E_C + E_P$
- The coils are thin parallels wires surrounding the facility.
  $\sum_{l=1}^{N_{coil}} if \mu_0 I_C \sqrt{\frac{r_0}{r}} \left[2 \frac{E_2(k_l)}{k_l} - E_1(k_l) \left(\frac{2}{k_l} - k_l \right) \right]$
- We use phasor notation.

Equations governing the plasma

$$\partial_t \rho + \nabla \cdot (\rho \mathbf{v}) = 0$$

$$\partial_t (\rho \mathbf{v}) + \nabla \cdot (\rho \mathbf{v}\mathbf{v}) = -\nabla p + \nabla \cdot \boldsymbol{\tau} + \mathbf{F}_L$$

$$\partial_t \left(\rho e + \frac{1}{2}\rho ||\mathbf{v}||^2 \right) + \nabla \cdot \left(\rho e \mathbf{v} + \frac{1}{2}\rho ||\mathbf{v}||^2 \mathbf{v} + p\mathbf{v} \right) = \nabla \cdot (\boldsymbol{\tau}\mathbf{v}) + \nabla \cdot \mathbf{q} + P_J$$

$$\boldsymbol{\tau} = \eta \left(\nabla \mathbf{v} + \nabla \mathbf{v}^T \right) - \frac{2}{3}\eta \nabla \cdot \mathbf{v}$$

$$\mathbf{q} = -\lambda \nabla T$$

$$F_z^L = \frac{\sigma_e}{4\pi f} \left[E_l^{Im} \partial_z E_l^{Re} - E_l^{Re} \partial_z E_l^{Im} \right]$$

$$F_r^L = \frac{\sigma_e}{4\pi f} \left[E_l^{Im} \frac{1}{r} \partial_r (r E_l^{Re}) - E_l^{Re} \frac{1}{r} \partial_r (r E_l^{Im}) \right]$$

$$P^J = \frac{\sigma_e}{2} \left[(E_l^{Im})^2 + (E_l^{Re})^2 \right]$$

Equations governing the plasma

$$\partial_t \rho + \nabla \cdot (\rho \mathbf{v}) = 0$$

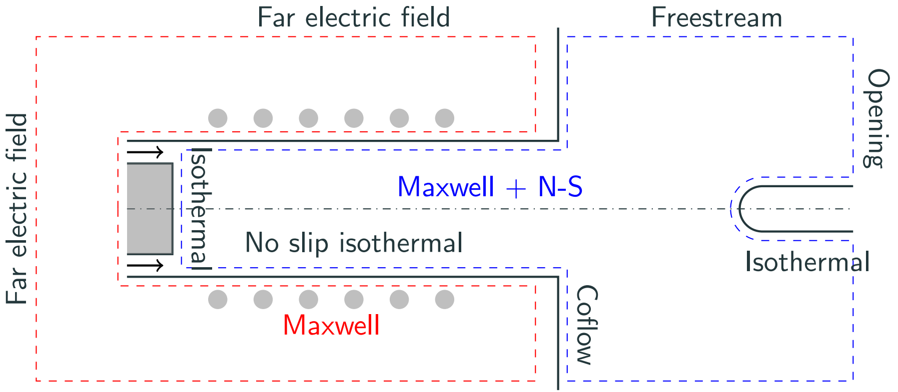
$$\partial_t (\rho \mathbf{v}) + \nabla \cdot (\rho \mathbf{v}\mathbf{v}) = -\nabla p + \nabla \cdot \boldsymbol{\tau} + \mathbf{F}_L$$

$$\partial_t \left(\rho e + \frac{1}{2}\rho||\mathbf{v}||^2\right) + \nabla \cdot \left(\rho e \mathbf{v} + \frac{1}{2}\rho||\mathbf{v}||^2 \mathbf{v} + p\mathbf{v}\right) = \nabla \cdot (\boldsymbol{\tau}\mathbf{v}) + \nabla \cdot \mathbf{q} + P_J$$

$$\partial^2_{zz} E_P + \frac{1}{r}\partial_r(r\partial_r E_P) - \frac{E_P}{r^2} = i2\pi f \mu_0 \sigma_e (E_C + E_P)$$

μ_0 : magnetic permeability, σ_e : electron electric conductivity, f : induction frequency

Computational domain and boundary conditions



Hybridized discontinuous Galerkin method

Conservative form of the equations

$$\begin{aligned}
\partial_t \rho \quad & + \nabla \cdot (\rho \mathbf{v}) & & = 0 \\
\partial_t (\rho \mathbf{v}) \quad & + \nabla \cdot (\rho \mathbf{v}\mathbf{v} + p\mathbb{I}) & - \nabla \cdot \boldsymbol{\tau} \quad & = \mathbf{F}_L \\
\partial_t (\rho E) \quad & + \nabla \cdot (\rho E \mathbf{v} + p\mathbf{v}) & - \nabla \cdot (\boldsymbol{\tau}\mathbf{v} + \mathbf{q}) \quad & = P_J \\
& & - \nabla \cdot (\nabla E_P) \quad & = -\frac{E_P}{r^2} - i2\pi f \mu_0 \sigma_e (E_C + E_P)
\end{aligned}$$

$$\partial_t \mathbf{w}(\mathbf{u}) + \nabla \cdot \mathbf{F}_c(\mathbf{u}) - \nabla \cdot \mathbf{F}_v(\mathbf{u}, \nabla \mathbf{u}) = \mathbf{S}(\mathbf{u}, \nabla \mathbf{u})$$

For N-S: $\mathbf{w} = \begin{pmatrix} \rho & \rho\mathbf{v} & \rho e + \rho\frac{||\mathbf{v}||^2}{2} \end{pmatrix}$, $\mathbf{u} = \begin{pmatrix} p & \mathbf{v} & T \end{pmatrix}$

Model problem and weak form

$$
\begin{aligned}
&\partial_t \mathbf{w}(\mathbf{u}) + \nabla \cdot (\mathbf{F}_c(\mathbf{u}) - \mathbf{F}_v(\mathbf{u}, \nabla \mathbf{u})) = \mathbf{S}(\mathbf{u}, \nabla \mathbf{u}) && \text{on } \Omega \\
&\mathbf{u} = \mathbf{u}_{bc} && \text{on } \partial\Omega_D \\
&\mathbf{F}_v \cdot \mathbf{n} = \mathbf{F}_{v,n,bc} && \text{on } \partial\Omega_N \\
&\mathbf{u}(t = 0) = \mathbf{U} && \text{on } \Omega
\end{aligned}
$$

$\mathbf{F}_{c,v}$: convective and diffusive fluxes.

$\mathbf{S}$: source terms.

$\mathbf{w}$: conservative variables.

$\mathbf{u}$, $\mathbf{u}_{bc}$: solution and boundary condition.

Ω : domain of the problem.

$\partial\Omega_{D,N}$: domain boundary for Dirichlet and Neumann BC.

$\mathbf{U}$: initial data.

Model problem and weak form

$$
\begin{aligned}
&\partial_t \mathbf{w}(\mathbf{u}) + \nabla \cdot (\mathbf{F}_c(\mathbf{u}) - \mathbf{F}_v(\mathbf{u}, \nabla \mathbf{u})) = \mathbf{S}(\mathbf{u}, \nabla \mathbf{u}) && \text{on } \Omega \\
&\mathbf{u} = \mathbf{u}_{bc} && \text{on } \partial\Omega_D \\
&\mathbf{F}_v \cdot \mathbf{n} = \mathbf{F}_{v,n,bc} && \text{on } \partial\Omega_N \\
&\mathbf{u}(t=0) = \mathbf{U} && \text{on } \Omega
\end{aligned}
$$

Weak form:

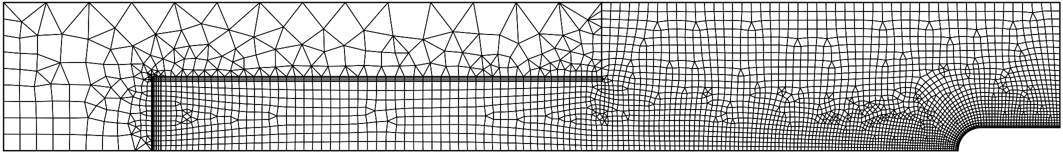
$$(\partial_t \mathbf{w} - \mathbf{S}, w)_\Omega - (\mathbf{F}_c - \mathbf{F}_v, \nabla w)_\Omega + \langle (\mathbf{F}_c - \mathbf{F}_v) \cdot \mathbf{n}, w \rangle_{\partial\Omega} = 0, \ \forall w \in L_2(\Omega)$$

with

$$(a, b)_K = \int_K ab \; dV, \quad \langle a, b \rangle_{\partial K} = \int_{\partial K} ab \; dS$$

Weak form: spatial discretization.

Division of Ω into elements (K):



$$\sum_K (\partial_t \mathbf{w} - \mathbf{S}, w)_K - (\mathbf{F}_c - \mathbf{F}_v, \nabla w)_K + \langle (\mathbf{F}_c - \mathbf{F}_v) \cdot \mathbf{n}, w \rangle_{\partial K} = 0,\ \forall w \in L_2(\Omega)$$

Weak form: Functional space reduction and solution

Restriction of functional space to a finite dimensional subspace of $L_2(\Omega)$. Ex:

$$w \in \mathcal{P}_p(K) \subset L_2(\Omega),$$

with $\mathcal{P}_p(K)$ polynomials of degree at most p on K.

If $\{\varphi_{K,i}\}$, $i = \{1, 2, ..., N_p\}$ is a basis (Lagrange, monomial, ...) of $\mathcal{P}_p(K)$, then

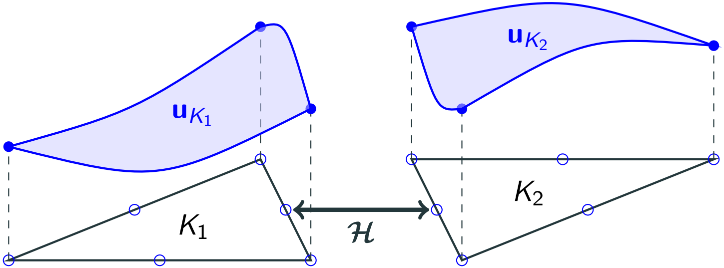
$$(\partial_t \mathbf{w} - \mathbf{S}, \varphi_{K,i})_K - (\mathbf{F}_c - \mathbf{F}_v, \nabla\varphi_{K,i})_K + \langle (\mathbf{F}_c - \mathbf{F}_v) \cdot \mathbf{n}, \varphi_{K,i} \rangle_{\partial K} = 0$$

u is then discretized as

$$\mathbf{u}_h = \sum_K \sum_{i=1}^{N_p} \mathbf{u}_{K,i} \varphi_{K,i}$$

Discontinuous Galerkin

$$(\partial_t \mathbf{w}_h - \mathbf{S}_h, \varphi_{K,i})_K - (\mathbf{F}_{c,h} - \mathbf{F}_{v,h}, \nabla \varphi_{K,i})_K + \langle \boldsymbol{\mathcal{H}}(\mathbf{u}_h, \nabla \mathbf{u}_h), \varphi_{K,i} \rangle_{\partial K} = 0$$



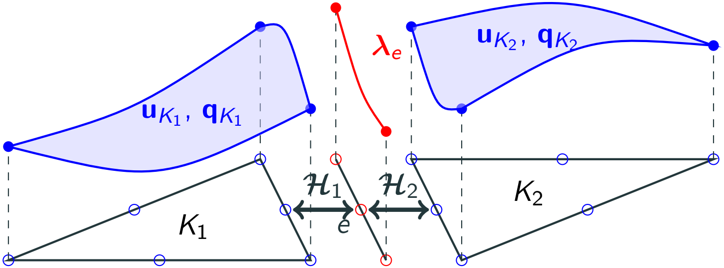
$$\mathbf{u}_h = \sum_K \sum_{i=1}^{N_p} \mathbf{u}_{K,i} \varphi_{K,i}$$

Hybridized Discontinuous Galerkin

$$(\partial_t \mathbf{w}_h - \mathbf{S}_h, \varphi_{K,i})_K - (\mathbf{F}_{c,h} - \mathbf{F}_{v,h}, \nabla \varphi_{K,i})_K + \langle \boldsymbol{\mathcal{H}}(\boldsymbol{\lambda}_h, \mathbf{u}_h, \mathbf{q}_h), \varphi_{K,i} \rangle_{\partial K} = 0$$

$$(\mathbf{q}_h, \boldsymbol{\tau}_{K,i})_K - (\mathbf{u}_h, \nabla \boldsymbol{\tau}_{K,i})_K + \langle \boldsymbol{\lambda}_h, \boldsymbol{\tau}_{K,i} \cdot \mathbf{n} \rangle_{\partial K_0} + \langle \mathbf{u}_{bc}, \boldsymbol{\tau}_{K,i} \cdot \mathbf{n} \rangle_{\partial K_{bc}} = 0$$

$$\langle [[\boldsymbol{\mathcal{H}}]], \mu_{e,i} \rangle_e = 0$$



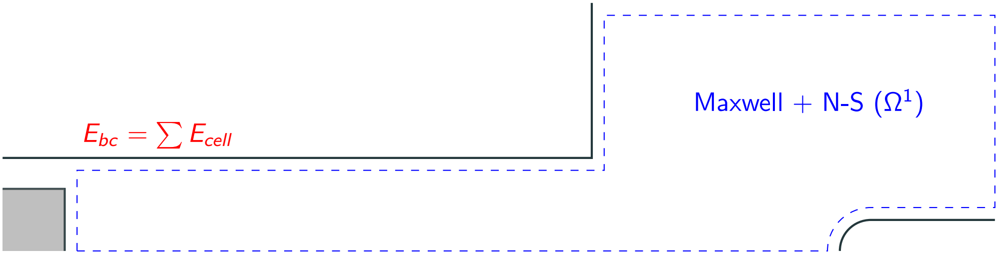
$$\mathbf{u}_h = \sum_K \sum_{i=1}^{N_p} \mathbf{u}_{K,i} \varphi_{K,i}$$

$$\mathbf{q}_h = \sum_K \sum_{i=1}^{N_p} \mathbf{q}_{K,i} \boldsymbol{\tau}_{K,i}$$

$$\boldsymbol{\lambda}_h = \sum_e \sum_{i=1}^{N_p} \boldsymbol{\lambda}_{e,i} \mu_{e,i}$$

ICP as a multi-domain problem

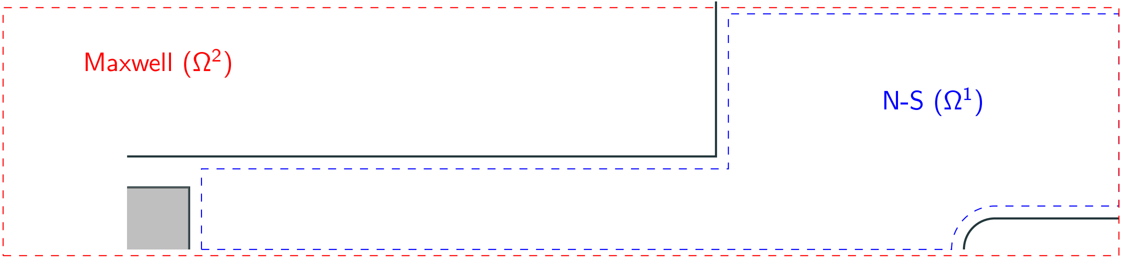
Single domain + integral boundary value



Compact but fills the jacobian matrix.

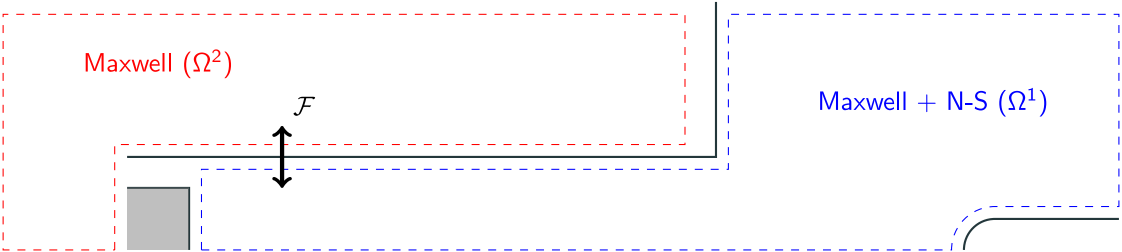
ICP as a multi-domain problem

Two overlapping domain



Most widely used. Solve N-S while Maxwell frozen, and *vice versa*. Slow convergence.

Multi-domain



Solves the system in a fully coupled manner, with two domains.

Modification of the model problem

$$
\begin{aligned}
&\partial_t \mathbf{w}^l(\mathbf{u}) + \nabla \cdot \Big(\mathbf{F}_c^l(\mathbf{u}) - \mathbf{F}_v^l(\mathbf{u}, \nabla \mathbf{u})\Big) = \mathbf{S}^l(\mathbf{u}, \nabla \mathbf{u}), && \text{on } \Omega^l \\
&\mathbf{u}^l = \mathbf{u}_{bc}^l, && \mathbf{x} \in \partial\Omega_d^l \\
&\mathbf{F}_v^l \cdot \mathbf{n} = \mathbf{F}_{v,n,bc}^l, && \text{on } \partial\Omega_n^l \\
&\mathcal{F}^{l,l'}(\mathbf{u}^l, \nabla \mathbf{u}^l, \mathbf{u}^{l'}, \nabla \mathbf{u}^{l'}) = 0, && \text{on } \gamma^{l,l'},\ l \neq l' \\
&\mathbf{u}^l(t=0) = \mathbf{U}^l, && \text{on } \Omega^l
\end{aligned}
$$

For ICP,

$$
\mathcal{F} = \begin{pmatrix} \nabla E_p^{Re,NS} \cdot \mathbf{n}^{NS} - \nabla E_p^{Re,MAX} \cdot \mathbf{n}^{MAX} \\ \nabla E_p^{Im,NS} \cdot \mathbf{n}^{NS} - \nabla E_p^{Im,MAX} \cdot \mathbf{n}^{MAX} \end{pmatrix}
$$

Modification of the discrete formulation

$$
\begin{aligned}
&\left(\partial_t \mathbf{w}_h - \mathbf{S}_h^K, \varphi_{K,i}\right)_K - \left(\mathbf{F}_{h,c}^K - \mathbf{F}_{h,v}^K, \nabla\varphi_{K,i}\right)_K + \left\langle \boldsymbol{\mathcal{H}}^K, \varphi_{K,i}\right\rangle_{\partial K} = 0\\
&(\mathbf{q}_K, \boldsymbol{\tau}_{K,i})_K - \left(\mathbf{u}_K, \nabla\boldsymbol{\tau}_{K',i}\right)_K + \langle \boldsymbol{\lambda}_{\partial K}, \boldsymbol{\tau}_{K,i}\cdot\mathbf{n}\rangle_{\partial K_0} + \langle \mathbf{u}_{bc}, \boldsymbol{\tau}_{K,i}\cdot\mathbf{n}\rangle_{\partial K_{bc}} = 0\\
&\langle [[\boldsymbol{\mathcal{H}}]], \mu_{e,i}\rangle_{e\setminus\bar{\Gamma}} + \langle \tilde{\mathcal{F}}^e, \mu_{e,i}\rangle_{e\cap\bar{\Gamma}} = 0
\end{aligned}
$$

with $\bar{\Gamma}$ the set of interfaces between subdomains.

Numerical fluxes : convection

Low mach requires to use preconditioned numerical flux

$$\boxed{\mathcal{H} = \mathcal{H}_c - \mathcal{H}_v}$$

$$\mathcal{H}_c = R\mathcal{H}_{c,n}(\mathbf{u}, \boldsymbol{\lambda}) = R(\dot{m} + \dot{m}_p)\Psi_\lambda - R|\dot{m}|(\Psi_\lambda - \Psi_U) + \mathbf{P}$$

with

$$\boldsymbol{\Psi} = \begin{pmatrix} 1 & v^\perp & v^\parallel & v_\theta & e + \frac{1}{2}||\mathbf{v}||^2 + \frac{p}{\rho} \end{pmatrix}^T$$

$$\mathbf{P} = \begin{pmatrix} 0 & p_\lambda & 0 & 0 \end{pmatrix}^T$$

$$\dot{m}_p = \frac{p_U - p_\lambda}{V_p}$$

Numerical fluxes : diffusion

$$\mathcal{H} = \mathcal{H}_c - \mathcal{H}_v$$

$$\mathcal{H}_v(\mathbf{u}, \lambda, \mathbf{q}) = \mathbf{F}_v(\boldsymbol{\lambda}, \mathbf{q}) \cdot \mathbf{n} + \gamma(h)(\boldsymbol{\lambda} - \mathbf{u})$$

with

$$\gamma = C \max_{f \in K} \left(\frac{1}{2\mathcal{V}} \sum_{f \in K} \mathcal{A} \right)$$

C depends on the diffusive transport coefficients.

Solution strategy: Newton solver with damping term

$$\boxed{\mathcal{N}(\mathbf{x}) = 0 \Rightarrow \mathcal{N}'(\mathbf{x}_k)\delta\mathbf{x}_k = -\mathcal{N}(\mathbf{x}_k)}$$

with damping term

$$\frac{\delta \mathbf{w}_h}{\tau_K} = \frac{\mathbf{w}(\mathbf{u}_{h,k+1}) - \mathbf{w}(\mathbf{u}_{h,k})}{\tau_{K,k}}$$

with

$$\tau_{K,k} = CFL_k \frac{h_K}{v_K} = CFL_k \frac{V_K}{\int_{\partial K} \lambda_{max} dS}$$

and the following CFL evolution law

$$CFL_k = \min\left(CFL_0 \left(\frac{||\mathcal{N}||_2^0}{||\mathcal{N}||_2^k} \right)^{\alpha}, CFL_{max} \right)$$

Matrix form of the system

$$\boxed{\mathcal{N}'(\mathbf{x}_k)\delta\mathbf{x}_k = -\mathcal{N}(\mathbf{x}_k)}$$

$$\underbrace{\begin{pmatrix} A & B & R \\ C & D & S \\ L & M & N \end{pmatrix}}_{\mathcal{N}'} \overbrace{\begin{pmatrix} \delta Q \\ \delta U \\ \delta \Lambda \end{pmatrix}}^{\delta \mathbf{x}} = \underbrace{\begin{pmatrix} F \\ G \\ H \end{pmatrix}}_{-\mathcal{N}} \Rightarrow \begin{cases} \underbrace{\begin{pmatrix} A & B \\ C & D \end{pmatrix}}_{\text{Block diagonal}} \begin{pmatrix} \delta Q \\ \delta U \end{pmatrix} = \begin{pmatrix} F \\ G \end{pmatrix} - \begin{pmatrix} R \\ S \end{pmatrix} \delta \Lambda \\ \begin{pmatrix} L & M \end{pmatrix} \begin{pmatrix} \delta Q \\ \delta U \end{pmatrix} + N \delta \Lambda = H \end{cases}$$

Matrix form of the system

$$\mathcal{N}'(\mathbf{x}_k)\delta\mathbf{x}_k = -\mathcal{N}(\mathbf{x}_k)$$

By eliminating δU and δQ, one gets the **global** system

$$\left(N - \begin{pmatrix} L & M \end{pmatrix} \begin{pmatrix} A & B \\ C & D \end{pmatrix}^{-1} \begin{pmatrix} R \\ S \end{pmatrix}\right) \delta\Lambda = H - \begin{pmatrix} L & M \end{pmatrix} \begin{pmatrix} A & B \\ C & D \end{pmatrix}^{-1} \begin{pmatrix} F \\ G \end{pmatrix}$$

System solved for $\boldsymbol{\lambda}$, then $\mathbf{u}$ and $\mathbf{q}$ are retrieved by small local matrices inversion.

$$\begin{pmatrix} \delta Q \\ \delta U \end{pmatrix} = \begin{pmatrix} A & B \\ C & D \end{pmatrix}^{-1} \begin{pmatrix} F \\ G \end{pmatrix} - \begin{pmatrix} A & B \\ C & D \end{pmatrix}^{-1} \begin{pmatrix} R \\ S \end{pmatrix} \delta\Lambda$$

The problem of power quenching

If the power dissipated is not maintained, the torch quenches!

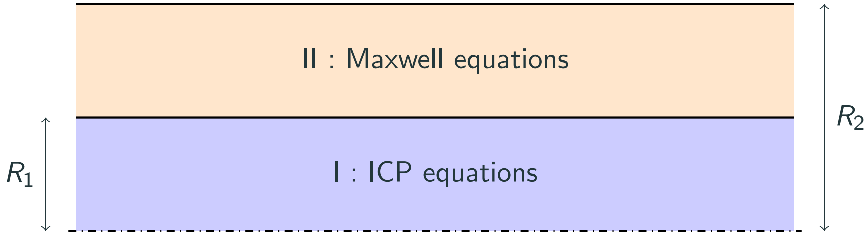
$$P_{tot} = \int_{NS} P_J dV$$

The current I_c is adapted throughout the simulation in order to reach a target power P_{target}.

$$I_c^{k+1} = I_c^k \sqrt{\frac{P_{target}}{P_{tot}^k}}$$

Results

Manufactured solution

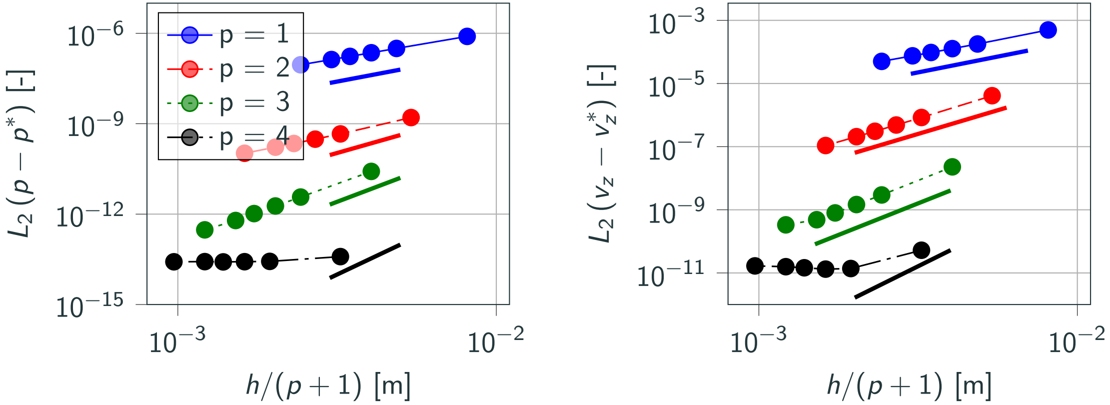


$$
\begin{aligned}
p &= \Delta p_0 f(z,r) + p_0 & T &= T_{min} + (T_{max} - T_{min})\, f(z,r) \\
v_z &= u_0 f(z,r) & E_p &= E_0 f(z,r)(1-i) \\
v_r &= u_0 f(z,r) & v_\theta &= u_0 f(z,r)
\end{aligned}
$$

with

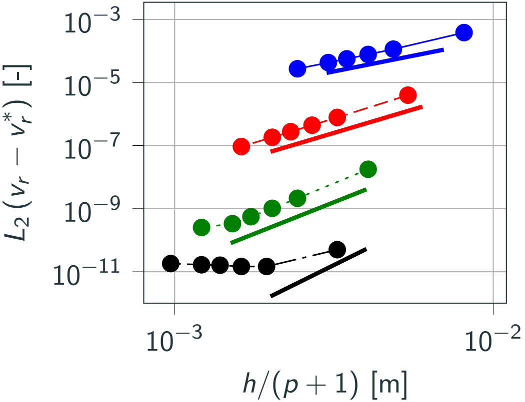
$$
f(z,r) = \left(\frac{rz}{RL}\right)^2 \exp\left(-\frac{z}{L} - \frac{r}{R_1}\right)
$$

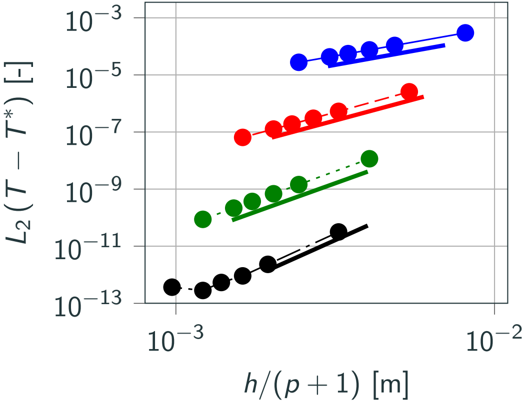
Convergence study : in the torch



Convergence order of $p+1$ is retrieved!

Convergence study : in the torch





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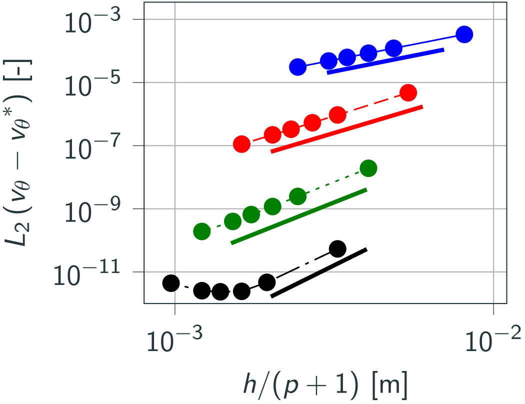

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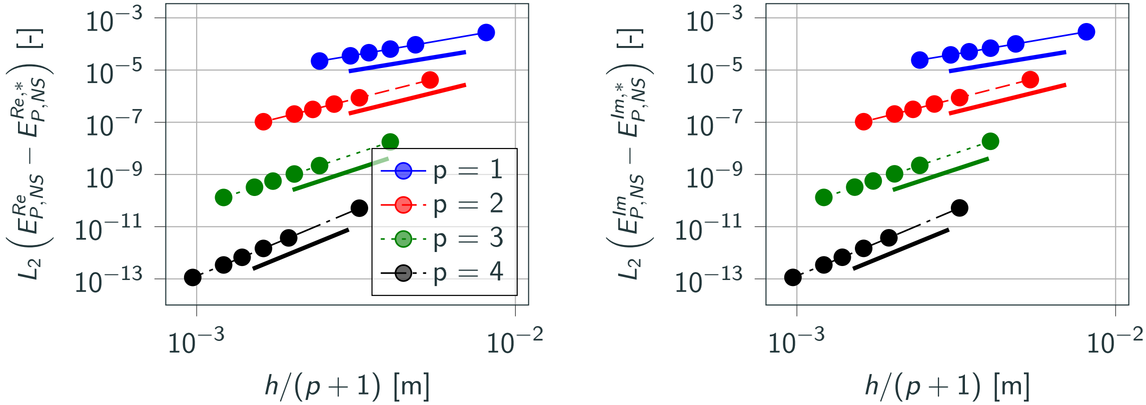

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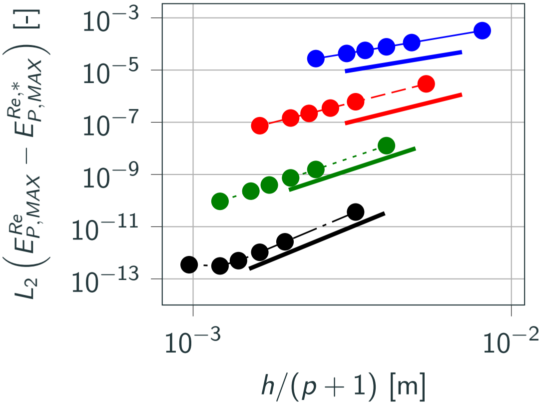
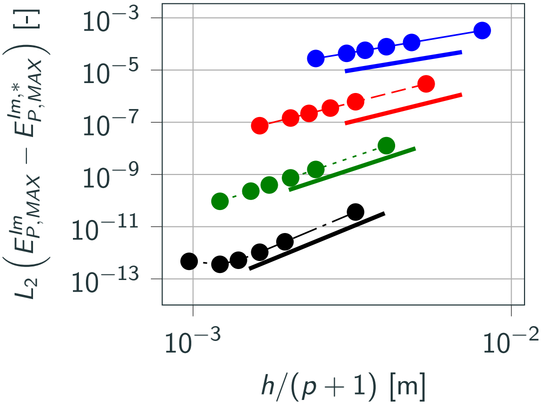
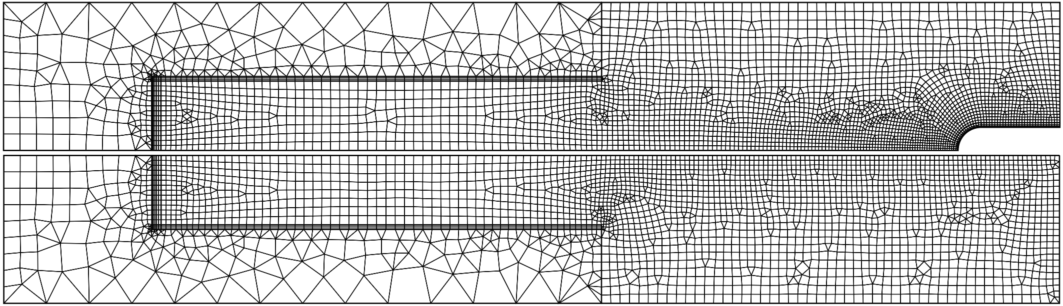




Convergence order of $p+1$ is retrieved!

Mesh and order dependence study



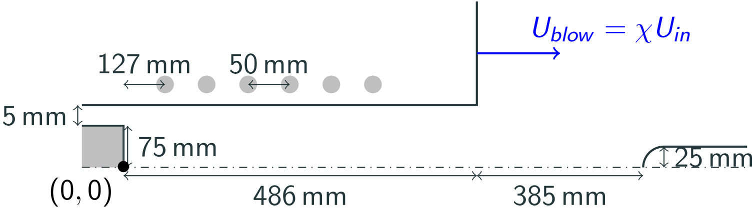
This mesh at $p = 4$ with 4013 elements for the probe case and 3055 for the freestream case is sufficient for capturing the solution accurately.

The mesh size close to the probe is 10 µm thick, while it is of 1 µm for FV!

The question of the coflow

A coflow is introduced in the chamber for obtaining a steady state, does it have an important impact?

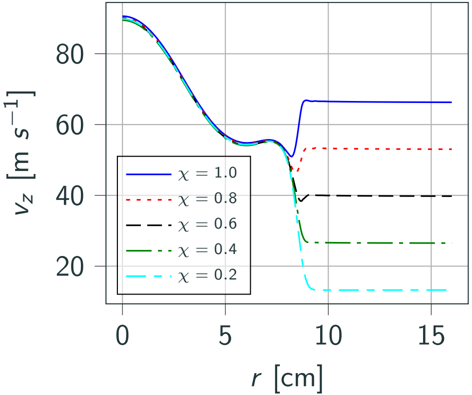
- $P = 100\,\text{kW}$
- $p_0 = 10\,000\,\text{Pa}$
- $f = 0.37\,\text{MHz}$
- $Q = 16\,\text{g s}^{-1}$
- $S = 20°$
- air (11 species)

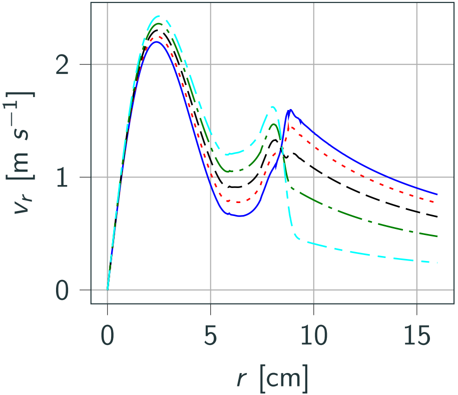


$$\chi = \frac{U_{blow}}{U_{in}} \in [0.2, 1.]$$

Effect of the coflow on the simulations

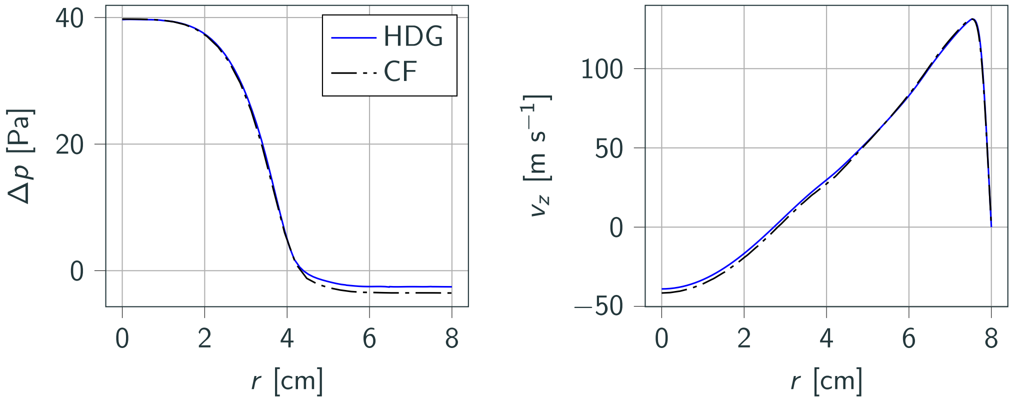
Only has an impact at the torch exit, outside the torch radius.





However, we were not able to obtain any steady state without coflow.

Comparison with the COOLFluiD solver (radial profile in the torch)



$p_0 = 5000\,\mathrm{Pa}$, $Q = 16\,\mathrm{g\,s^{-1}}$, $T_{wall} = T_{inlet} = 350\,\mathrm{K}$, $P = 100\,\mathrm{kW}$, $f = 0.37\,\mathrm{MHz}$, $S = 0$, air (11 species), $\chi_{HDG} = 0.2$.

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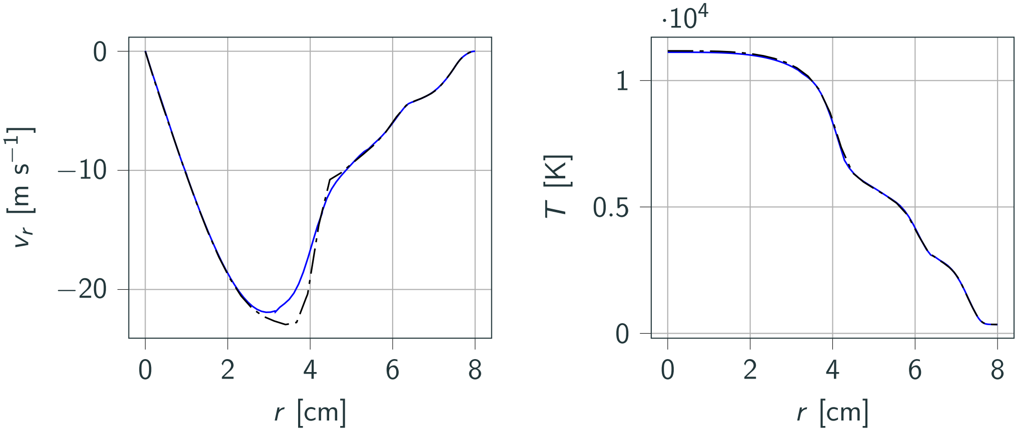

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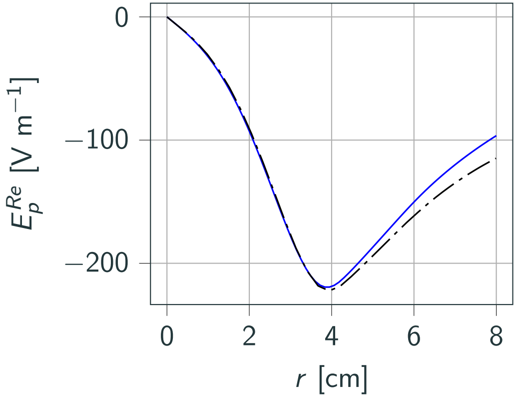

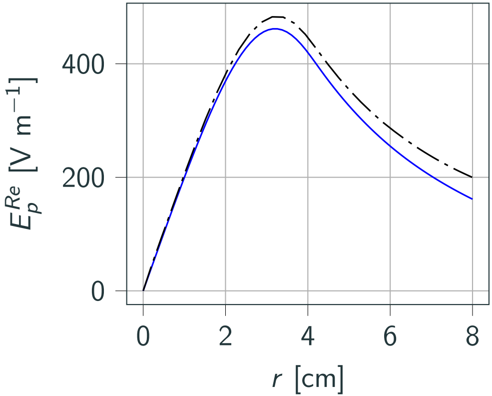

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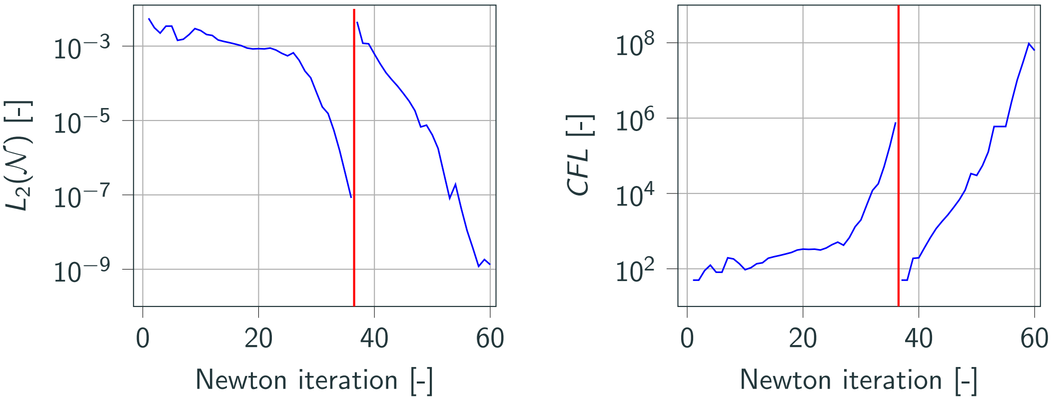
Convergence history

We use GMRES solver with at most 50 vectors and a ILU(4) preconditioner.

We first solve with polynomial degree 1 then 4 (red separation line).



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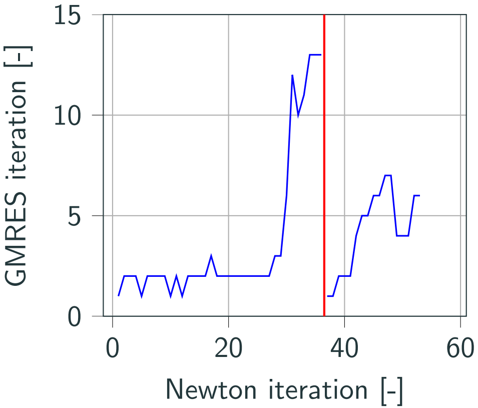


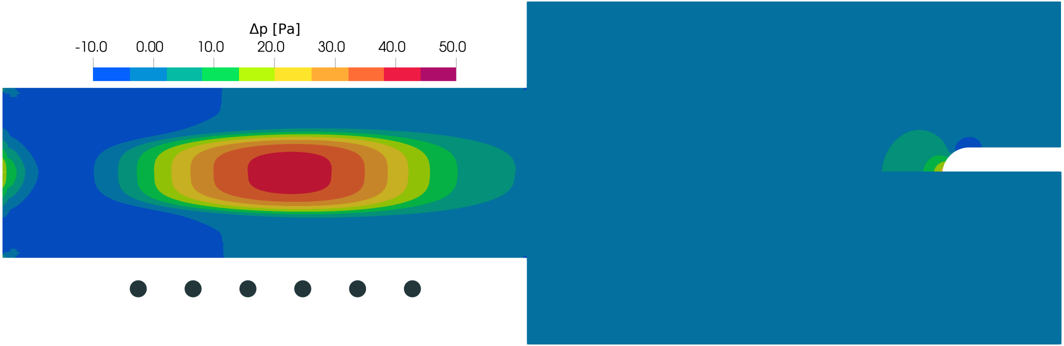
Fast and robust solver for ICP, able to produce meaningful results in 20 minutes of time on a single laptop using 8 OMP threads!

DOFS $\lambda \simeq 400\,000$

DOFs $w \simeq 1\,000\,000$

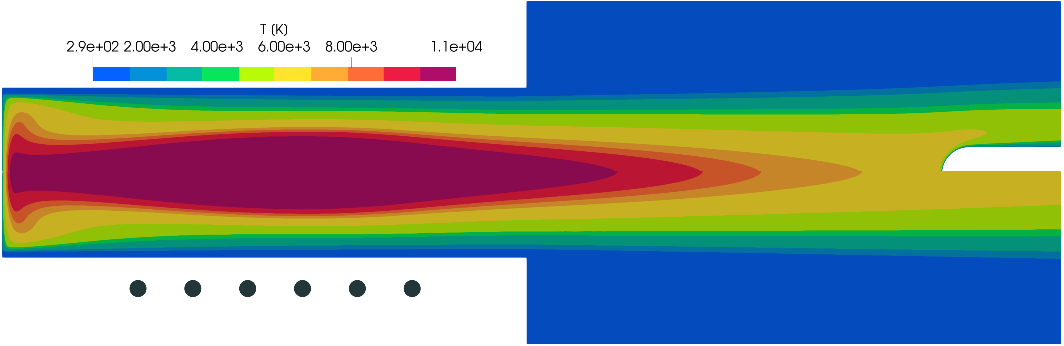
DOFS $q \simeq 2\,000\,000$

Flow fields in ICP: gauge pressure



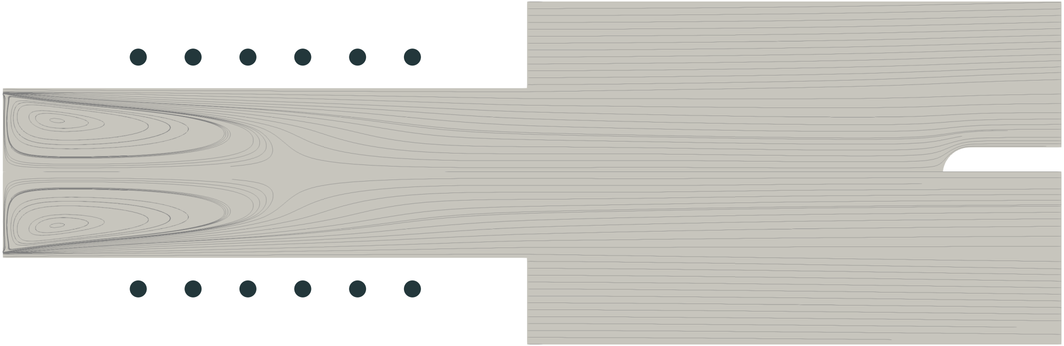
The pressure is almost constant everywhere ($p_0 = 10\,000\,\mathrm{Pa}$).

Flow fields in ICP: temperature



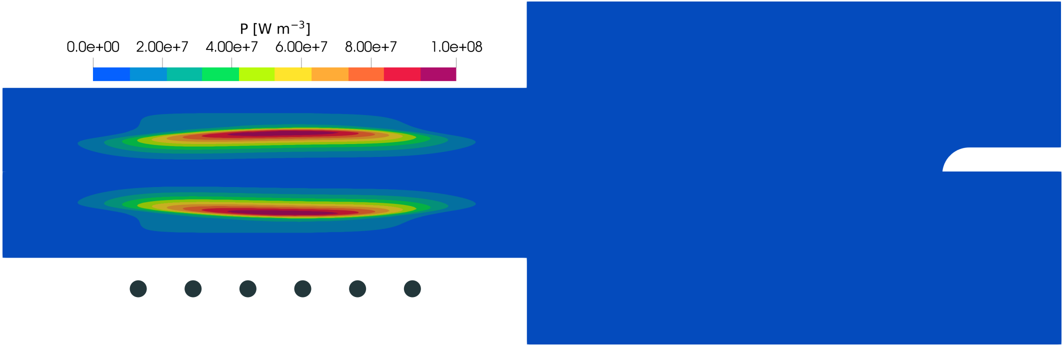
Large temperature gradients.

Flow fields in ICP: streamlines



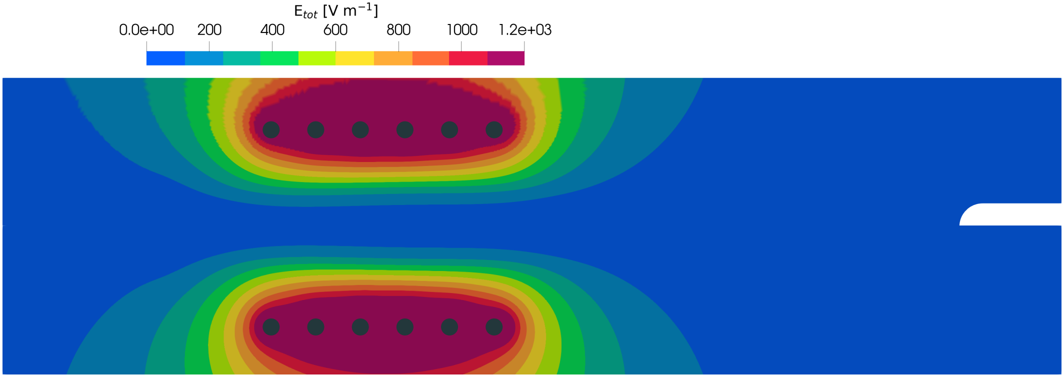
$$\nabla \cdot \mathbf{v} \simeq 0$$

Flow fields in ICP: volume power

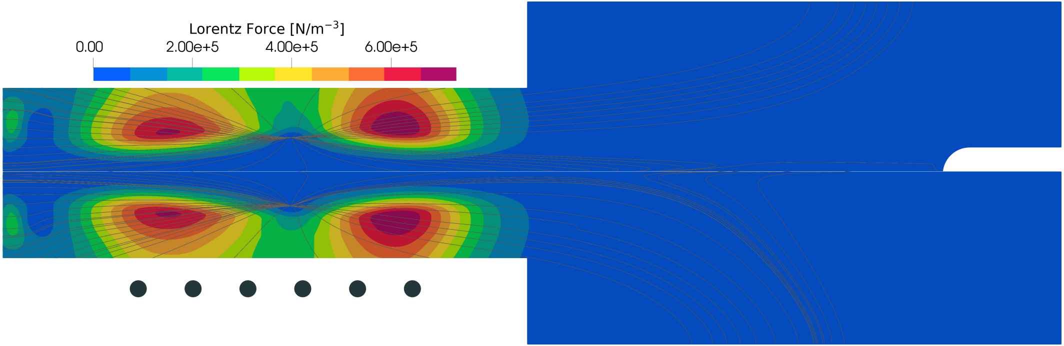


Volume power peak off-axis, where the coupling is the strongest.

Flow fields in ICP: Total electric field



Flow fields in ICP: Lorentz force



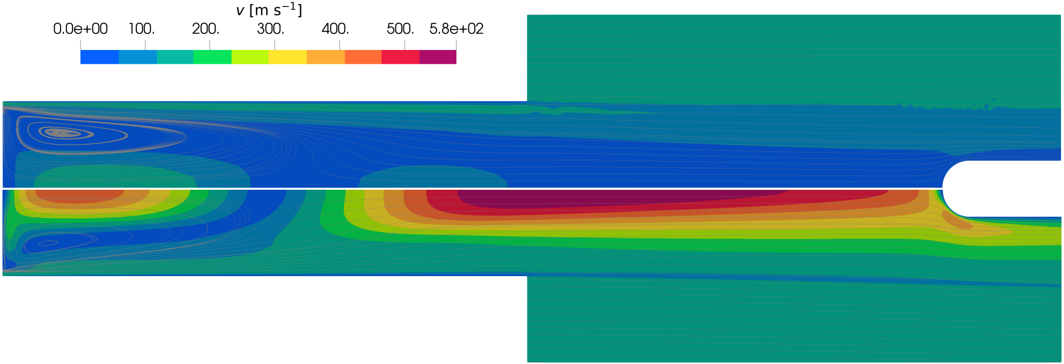
Parametric study

We performed a parametric study of ICP for the parameters used by the experimenters.

- Q from $8\,\mathrm{g\,s^{-1}}$ to $16\,\mathrm{g\,s^{-1}}$.
- p_0 from $5000\,\mathrm{Pa}$ to $20\,000\,\mathrm{Pa}$.
- P_{tot} from $50\,\mathrm{kW}$ to $200\,\mathrm{kW}$.
- f from $10\,\mathrm{kHz}$, to $1\,\mathrm{MHz}$.
- S from 0° to 20°.

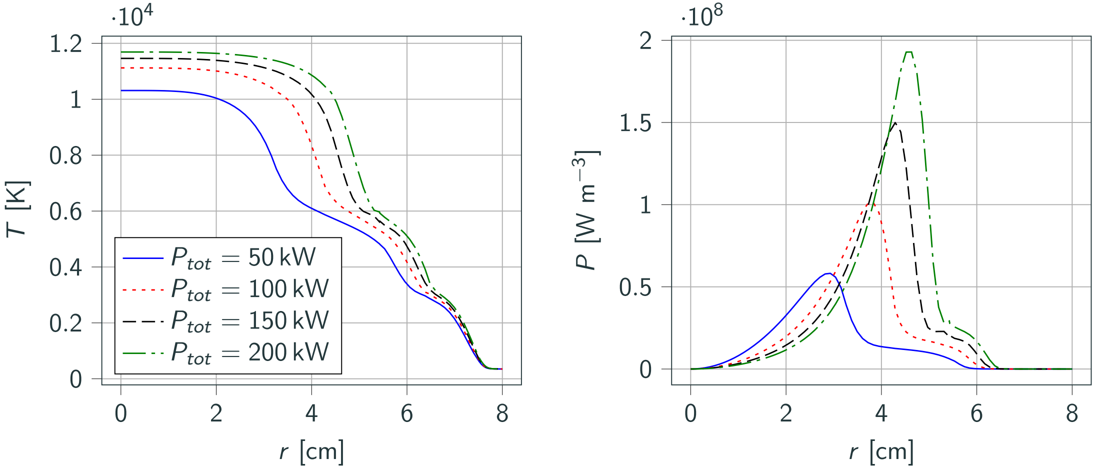
The complete discussion is in the thesis.

Power dissipated in the facility: impact on mass flow rate

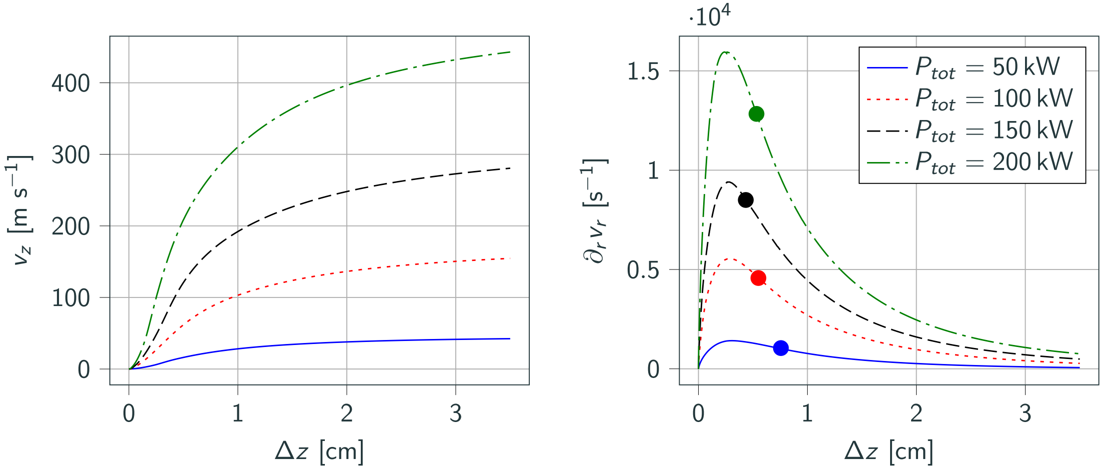


$P_{tot} = 50\,\text{kW}$ (top) and $P_{tot} = 200\,\text{kW}$ (bottom)

Power dissipated in the facility: impact in the torch



Power dissipated in the facility: impact along the stagnation line



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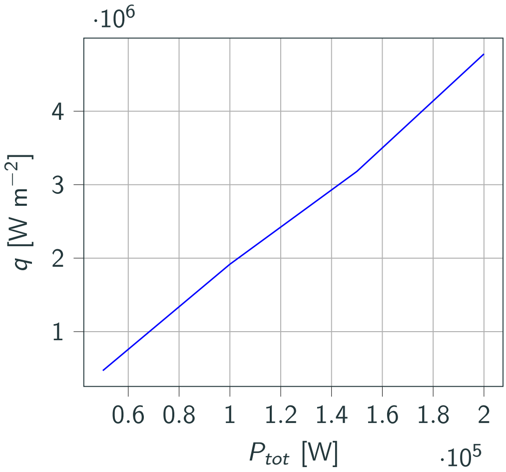

Conclusions and future work

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Yes, but there is still some parameter tuning.

However, much easier to use and give meaningful results fast.

A lot of simulations can be performed "eyes closed".

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Conclusions

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Q3 Can the new solver be easily adapted to the new experiments performed in ICP facilities?

We believe so, but there is still much work to do...

Going 2D, 3D, and complex chemistry and radiation for unsteady simulations will need much more computational power.

For supersonic, shock capturing and redesign of the numerical flux.

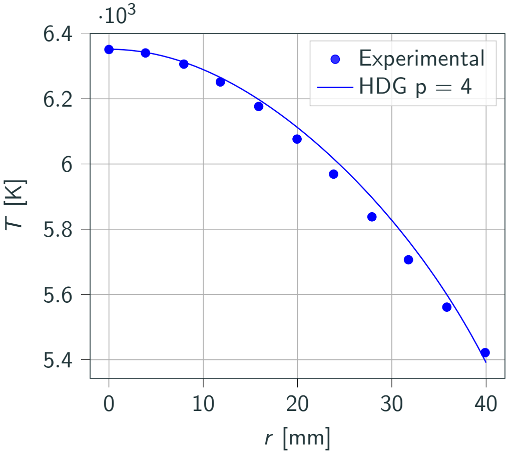
Future work

Main effort is to put our developments in the ARGO DG solver, which is fully paralleled and equipped shock capturing technologies. (Landry Riou)

We see the following roadmap:

- Going 3D.
- Going unsteady (also relieving the average over the induction current period).
- Development of chemistry (non equilibrium).
- Development of radiation.
- For all of the above, comparison with experiment when possible.

Comparison with experiment

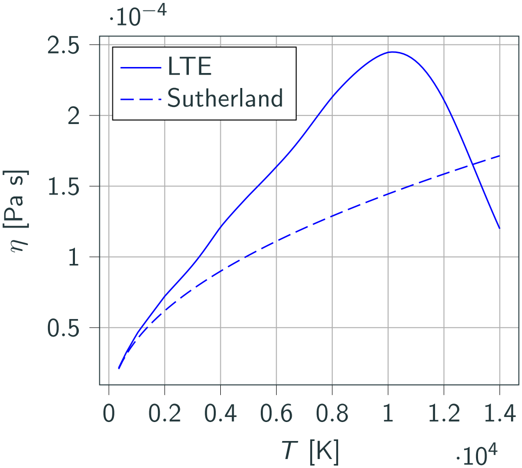


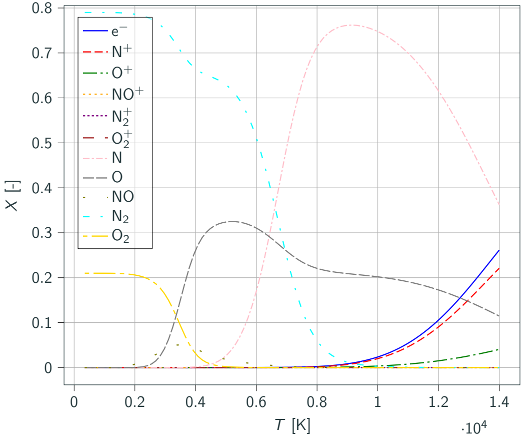
- $p_0 = 200\,\mathrm{mbar}$
- $Q = 16\,\mathrm{g\,s^{-1}}$
- $P_{exp.} = 200\,\mathrm{kW}$
- $P_{num.} = 78\,\mathrm{kW}$
- Freestream, no swirl, air.

Comparison of the experimental $T(r)$ at 385 mm from the torch exit, from $r = 0$ to 40 mm.

Backup

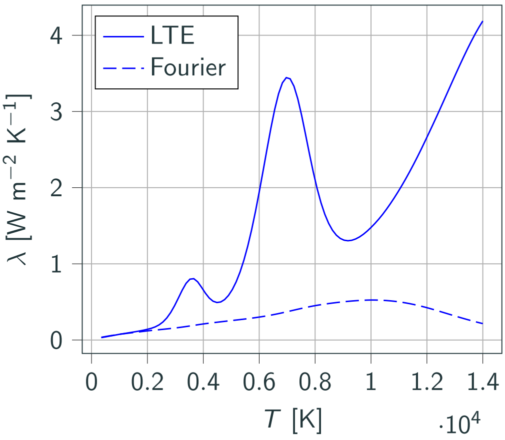
Transport properties

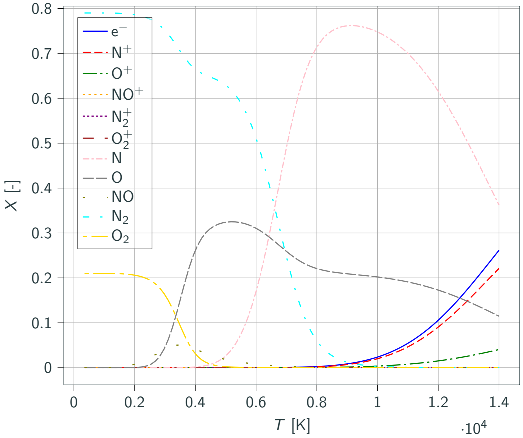




$$\eta \propto \sqrt{T}/\sigma$$

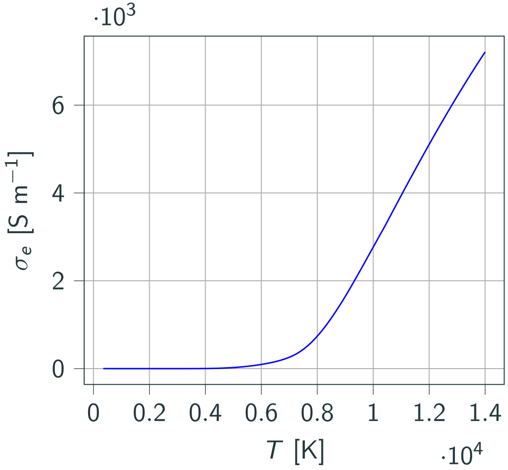
Transport properties

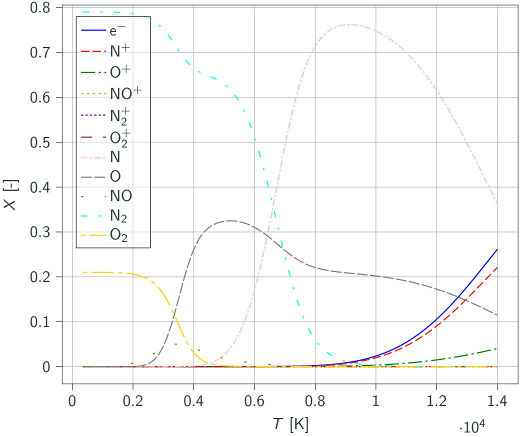




$$\lambda_r \propto \frac{dX}{dT}$$

Transport properties





$$\sigma_e \propto X_e$$

Plasma governing equations : the Boltzmann equation

$$\mathcal{D}_i(f_{i,I}) = \partial_t f_{i,I} + \mathbf{c}_i \cdot \nabla f_{i,I} + \frac{\mathbf{F}_i}{m_i} \cdot \partial_{\mathbf{c}_i} f_{i,I} = \mathcal{S}_i + \mathcal{C}_i$$

$f_i(\mathbf{x}, \mathbf{c})$: probability density function.

$\mathcal{S}_i$: scattering collision operator (non chemically reacting).

$\mathcal{C}_i$: chemically reacting collision operator.

Provided $Kn = \frac{\lambda}{L} \ll 1$ (continuous medium)

$$\rho = \sum_i \int m_i f_i d\mathbf{c}_i. \quad \rho\mathbf{v} = \sum_i \int m_i \mathbf{c}_i f_i d\mathbf{c}_i, \ldots$$

Solution strategy

The HDG system is solved at steady-state such that

$$\left(\cancel{\partial_t \mathbf{w}_h} - \mathbf{S}_h^K, \varphi_{K,i}\right)_K - \left(\mathbf{F}_{h,c}^K - \mathbf{F}_{h,v}^K, \nabla\varphi_{K,i}\right)_K + \left\langle \boldsymbol{\mathcal{H}}^K, \varphi_{K,i}\right\rangle_{\partial K} = 0$$
$$\left(\mathbf{q}_K, \boldsymbol{\tau}_{K,i}\right)_K - \left(\mathbf{u}_K, \nabla\boldsymbol{\tau}_{K',i}\right)_K + \langle \boldsymbol{\lambda}_{\partial K}, \boldsymbol{\tau}_{K,i}\cdot\mathbf{n}\rangle_{\partial K_0} + \langle \mathbf{u}_{bc}, \boldsymbol{\tau}_{K,i}\cdot\mathbf{n}\rangle_{\partial K_{bc}} = 0$$
$$\langle [[\boldsymbol{\mathcal{H}}]], \mu_{e,i}\rangle_{e\setminus\bar{\Gamma}} + \langle \tilde{\mathcal{F}}^e, \mu_{e,i}\rangle_{e\cap\bar{\Gamma}} = 0$$

If the system is written as $\mathcal{N} = 0$, then the Newton method solves

$$\mathcal{N}'(\mathbf{x}_k)\delta\mathbf{x}_k = -\mathcal{N}(\mathbf{x}_k)$$

with $\delta\mathbf{x}_k = \mathbf{x}_{k+1} - \mathbf{x}_k$.

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The HDG system is solved at steady-state such that

$$\left(\frac{\delta \mathbf{w}_h^K}{\tau_{K,k}} - \mathbf{S}_h^K, \varphi_{K,i}\right)_K - \left(\mathbf{F}_{h,c}^K - \mathbf{F}_{h,v}^K, \nabla\varphi_{K,i}\right)_K + \left\langle \boldsymbol{\mathcal{H}}^K, \varphi_{K,i}\right\rangle_{\partial K} = 0$$

$$(\mathbf{q}_K, \boldsymbol{\tau}_{K,i})_K - \left(\mathbf{u}_K, \nabla\boldsymbol{\tau}_{K',i}\right)_K + \langle \boldsymbol{\lambda}_{\partial K}, \boldsymbol{\tau}_{K,i}\cdot\mathbf{n}\rangle_{\partial K_0} + \langle \mathbf{u}_{bc}, \boldsymbol{\tau}_{K,i}\cdot\mathbf{n}\rangle_{\partial K_{bc}} = 0$$

$$\langle [[\boldsymbol{\mathcal{H}}]], \mu_{e,i}\rangle_{e\setminus\bar{\Gamma}} + \langle \tilde{\mathcal{F}}^e, \mu_{e,i}\rangle_{e\cap\bar{\Gamma}} = 0$$

If the system is written as $\tilde{\mathcal{N}} = 0$, then the Newton method solves

$$\tilde{\mathcal{N}}'(\mathbf{x}_k)\delta\mathbf{x} = -\tilde{\mathcal{N}}(\mathbf{x}_k)$$

with $\delta\mathbf{x} = \mathbf{x}_{k+1} - \mathbf{x}_k$.

In order to stabilize the solver, a damping term is added.